

Building the Cell Back

Three ways to put hemoglobin in a package again, and what each one gets right that a bare polymer does not

Every hemoglobin-based oxygen carrier that has reached a clinic is a bare protein. Whether the protein is cross-linked, polymerized, or wrapped in polymer chains, it floats free in plasma with nothing between it and the vessel wall. That design carries the direct exposure that produced the troubles of the class, and a persistent minority of researchers has argued that the way forward is to take a step back: put the hemoglobin in a package again.

A red cell is not a bag of hemoglobin. It is a machine. A membrane keeps the protein in and the plasma out, a set of enzymes keeps the iron reduced, a bath of small molecules tunes oxygen affinity, and a flexible skeleton lets the whole thing fold through a capillary narrower than itself. Recreating any of that outside the body is difficult. Three programs, each aiming at a different piece of the machine, have advanced far enough to teach us something. This essay describes them, and it puts them beside the bovine polymers to ask a fair question: what is a packaged carrier for?

The case for a membrane

Start with the logic of the package, because it is the same logic that evolution followed. A membrane does four jobs for a red cell.

It keeps hemoglobin away from the vessel wall. Free hemoglobin scavenges nitric oxide, a gas the wall uses as a relaxing signal. A cell keeps the scavenger inside, at a distance.

It keeps hemoglobin away from the kidney. A protein small enough to leak through the filter is lost, and can be toxic there. A package that is much larger than the filter's pores cannot leak.

It supports a chemical environment. Inside a cell, enzymes and small molecules keep iron in the working form and regulate affinity. Inside a synthetic package, at least some of those functions can be included by adding the right compounds.

And it controls the concentration. A packaged carrier can hold hemoglobin at the same high concentration as a red cell, so the oxygen capacity per particle is large, while the fluid in which it travels stays thin.

These are real advantages. They have to be weighed against the costs of a package: it must be made, it must be stable, and it must not trigger the immune system or clog the very vessels it hopes to reach.

Program one: the hemoglobin vesicle

The most advanced packaged carrier is the hemoglobin vesicle, developed over decades by research groups in Japan. It consists of a highly concentrated hemoglobin solution enclosed in a bilayer of phospholipids, the same class of molecule that forms a cell membrane. The particles are small spheres about a quarter of a micrometer across, roughly thirty times smaller than a

red cell. The surface is decorated with polyethylene glycol to reduce recognition by the immune system and to prevent the particles from sticking to one another.

The hemoglobin inside is purified from outdated donated human red cells, and the vesicles are formulated with additional compounds that mimic the small-molecule environment of a cell, including a molecule to tune oxygen affinity and reducing agents to protect the iron. The developers have reported that the product can be stored for long periods, and have described it as a candidate for emergency use where blood is unavailable. A first-in-human phase 1 trial was published in 2022 in a hematology journal. It enrolled a small group of healthy volunteers, gave them single doses, and reported on safety and tolerability. I have not been able to confirm its detailed results, and I would treat any claim about its performance in patients as unproven.

The design has a few clear attractions. It is built from human hemoglobin and lipids, so it introduces no animal proteins. The particle size is well above the range that escapes the kidney. The iron is protected inside a packaged environment. And its behavior in animals has been extensively studied, including the observation, reported in rodents, that injection did not cause the anaphylactic reactions that some lipid-based particles are known to provoke.

The design has clear drawbacks too. The raw material is human blood, so it depends on the same donor supply it hopes to supplement, although it is made from expired units that would otherwise be discarded. Lipid particles are cleared by the reticuloendothelial system, the same scavenger cells that take up perfluorocarbon emulsions, and this clearance limits the half-life and raises questions about long-term effects on those cells. And manufacturing a uniform lipid particle at scale is a specialized process.

Program two: synthetic cells from the ground up

A second approach attempts to build the package from synthetic materials, with no cells and no donor blood in the loop. One design in this family uses hemoglobin held in a nanoparticle with a polymer shell, together with molecules that mimic the enzymes and affinity regulators of the red cell, and a formulation that can be freeze-dried into a powder. The stated goal is a product that can be stored at room temperature and reconstituted with water in the field.

I want to be very careful about how I describe this. It is a research program from a small number of laboratories, with published animal data and a strong narrative but no completed large human trial that I could verify. I mention it because it shows the direction in which the idea is being pushed: from a natural package to an engineered one, with each function of the red cell replaced by a designed component. The appeal is obvious. If a synthetic package works, it removes the dependence on blood entirely. The uncertainty is equally obvious. Each added component adds an interaction to be tested, and complex particles have a history of surprising behavior in the body.

Here is a way to think about the difference between the two programs. The vesicle is a conservative design: it keeps nature's blueprint, the phospholipid bilayer, and works to stabilize it. The synthetic nanoparticle is a radical design: it keeps the function and replaces the structure. Conservative designs tend to fail slowly and modestly. Radical designs tend to succeed dramatically or fail quietly. Neither can be judged in advance.

Program three: growing the real thing

The third approach avoids substitutes entirely. Instead of building a package, grow a genuine red cell in the laboratory. Blood stem cells, taken from donors, can be coaxed in culture to mature into red cells, which are then harvested and transfused.

A study in the United Kingdom took this beyond the bench. In a trial conducted in the early 2020s, healthy volunteers received small volumes of laboratory-grown red cells that had been tagged so their survival could be measured against the volunteers' own cells. The aim was not therapy but a demonstration that the cells could circulate and survive in a human. Reports described it as the first trial of its kind in the world, and the results were being analyzed at the time I gathered this material. I cannot give a verified account of the findings, and I will not guess.

The theoretical attraction is immense. A cultured cell has the membrane, the enzymes, the affinity regulation and the flexibility, because it is a red cell. If cultured cells could be made in volume, they could supply patients whose blood types are rare or who have developed antibodies against common types, a group for whom compatible blood is hard to find.

The practical obstacle is scale and cost. A single unit of red cells contains on the order of a trillion cells. Producing that number in culture, with the nutrients and growth factors required, and doing so sterilely and reproducibly, is orders of magnitude beyond what current laboratory methods deliver. The realistic near-term use is small-volume, high-value: rare-type patients, not trauma. For emergency oxygen, a cultured cell is not a competitor to a shelf-stable bag. It is a complementary tool for a different job.

The arithmetic of a trillion cells

Scale deserves a worked example, because it is the point at which many elegant ideas quietly die. The numbers here are round figures for illustration.

A unit of donated red cells, roughly 300 milliliters, contains on the order of two trillion cells. A single bleeding adult may need several units, so the raw count for one patient runs to ten trillion or more. Now suppose a culture system can expand one starting stem cell into a hundred thousand mature red cells, a generous assumption. To make one unit, you would need twenty million starting cells. To make a hundred units, two billion. Those are numbers a bioreactor can handle in principle, but each of those cells needs weeks of nutrients, growth factors and sterile handling, and the cost of the growth factors alone can run into thousands of dollars per unit at laboratory prices.

The point is not that the arithmetic makes the idea impossible. It is that it sorts applications by value. A unit that costs thousands of dollars to grow makes sense for a patient with a rare blood type who would otherwise go untreated. It makes no sense for a trauma patient who needs six units in an hour. That is why I placed cultured cells beside the shelf-stable carriers as a complement and not a competitor.

The same reasoning applies to vesicles and synthetic particles, in a different form. Their cost is not in growth factors but in purified hemoglobin, in lipids of pharmaceutical grade, and in the yield of a multi-step manufacturing process. A packaged carrier that costs many times as much

as a bare polymer per gram of hemoglobin has to earn the difference through a benefit that patients and payers can see.

What an animal study cannot tell you about a package

Packaged carriers pose a particular challenge for preclinical testing. The reason is that many of their problems depend on the immune system and on the cells that clear particles, and these differ between species.

Consider complement activation. Some lipid and polymer particles trigger a chain of proteins in the blood that can cause flushing, chest tightness, breathing difficulty and a drop in blood pressure within minutes of the first infusion. The reaction does not require prior exposure, and it does not involve the antibodies of classical allergy. Its strength varies enormously among species. Pigs are notoriously sensitive, and rodents are far less so. A finding of no reaction in mice is therefore reassuring only up to a point.

Consider clearance. The scavenger cells that take up particles are more or less abundant in different species, and their capacity to handle repeated doses is uncertain. A single injection in a rodent cannot tell you what a series of injections in a person will do.

Consider size and shape. A particle that passes easily through a rodent capillary may behave differently in a human vessel, and rigid particles may be filtered by the lung.

The upshot is that the first human doses are a genuinely informative experiment, more so than for a simple protein, and the design of that experiment deserves care: small starting doses, slow infusion, monitoring of complement markers and of inflammatory signals, and staged escalation.

The ethics of the first vial

The mention of healthy volunteers in the vesicle and cultured cell trials invites a question worth asking. Why test an emergency product in people who are not in an emergency?

The reason is safety. Before a carrier is given to a critically ill patient, in whom side effects are hard to separate from the disease, it is given to healthy people so that its effects can be seen cleanly. This is standard practice for new drugs. But an oxygen carrier is a particular case, because its most likely side effects, changes in blood pressure, in organ function and in laboratory tests, are exactly the ones that a healthy volunteer will experience without benefit.

The ethics therefore turn on transparency and on choosing an appropriate dose. Volunteers must be told clearly that they will receive an unapproved product, that it will not help them, and that measurements will be taken to look for harm. The doses should be low enough to make serious injury very unlikely and high enough to be informative. Some carriers turn the urine or plasma a strange color, and volunteers should be prepared for that. And follow-up should last long enough to catch delayed effects, particularly for products that are cleared slowly by scavenger cells.

There is a further consideration that applies to patient advocacy. When a foundation supports a trial, it should ask who is being enrolled, how they are paid, and whether the payment is proportionate. A volunteer trial that recruits from vulnerable groups because they need money

is ethically different from one that recruits broadly, and the difference shows up in the quality of the consent.

What each package does to the hemoglobin problem

It helps to line the three programs up beside a bovine polymer, on the problems that the whole field cares about.

On nitric oxide scavenging, the package helps: a membrane keeps the protein away from the wall. The polymer relies on size and chemistry to reduce the effect, and only partly succeeds. The cultured cell solves it completely.

On kidney leakage, all the packaged particles are large enough to avoid filtration. The polymer manages by raising its average size, with the small end of the distribution as a lingering risk.

On iron oxidation, the package can carry protective enzymes and reducing agents. The polymer is exposed and depends on the manufacturing chemistry and on storage conditions.

On immune recognition, the vesicle and the nanoparticle use polyethylene glycol to reduce it, while the bovine polymer relies on the protein's chemistry and on the fact that hemoglobin is a familiar molecule. The cultured cell is the most natural but requires blood group matching.

On shelf life and convenience, the vesicle and the freeze-dried nanoparticle aim at room temperature. The polymer already achieves it. The cultured cell is the least convenient.

On maturity, the polymer leads by decades of human data. The vesicle is at the start of human testing. The nanoparticle and the cultured cell are earlier still.

That last line is the crux. The bare polymer is the least elegant design and the most tested one. It has been given to thousands of patients, manufactured at scale, sold in three markets, and used in emergencies. The elegant packages have not.

Why the bovine polymer still matters

I want to resist the temptation to conclude that a package is simply better. It may prove to be, in the long run. But a patient reading this in the near term should recognize two things.

First, the packaged carriers inherit a new set of problems, which are not zero. The immune system is especially wary of lipid and polymer particles, and there is a known reaction to some such particles, in which complement activation produces a sudden, sometimes dangerous response on first exposure. Animal studies with hemoglobin vesicles have found no anaphylactic reactions in rodents, which is encouraging but does not settle the question in humans. The reticuloendothelial system may be burdened by repeated doses of particles. The long-term fate of the lipids is not fully described.

Second, the timeline is different. A bovine polymer exists. It can be manufactured today and was administered to patients on a compassionate basis. A packaged carrier is a development program whose completion is measured in years. For a patient who needs an option now, the difference is decisive.

There is a reasonable middle view. The bare polymer may be the bridge. It occupies the space until something better arrives, and the experience gained with it, in manufacturing, in regulation, in trial design, in measurement, in the ethics of emergency access, will be inherited by whatever succeeds it. Every technology in this essay will draw on the path that the polymer cleared, or failed to clear.

What to watch

For a reader who wants to follow this field with a critical eye, here are the markers I would track.

For the vesicle: results of larger human trials, data on repeat dosing, and the identity of a manufacturer with the capacity to produce at scale.

For the synthetic nanoparticle: published human data of any kind, and independent replication of the animal results.

For cultured cells: the yield per donor cell, the cost per unit, and the number of patients treated outside a trial.

For the polymer: the outcome of any new trial, the continuity of manufacturing and the pattern of adverse events in expanded access.

None of these will arrive quickly. All of them will be visible in the public record if someone is paying attention.

Closing thought

Evolution solved the oxygen problem by putting hemoglobin in a cell, and most of what has gone wrong in the history of oxygen carriers can be read as the consequences of taking it out. The three programs here attempt to put it back in some form. It is entirely possible that the future of the field lies in a package, and equally possible that the bare polymer, refined and used in the right niche, will serve for decades. The prudent position for patients is to support the science that tests both, and to insist that each be judged on the same terms: what it does, in whom, compared with what.

Further reading

The hemoglobin vesicle work is described in reviews by its developers, including a 2022 article on storable, ready-to-use artificial red blood cells for emergency medicine and a first-in-human phase 1 report in *Blood Advances* the same year. A study of hemoglobin vesicle injection in rodents reported an absence of anaphylactic reactions. Programs on synthetic and cultured red cells are described in the recent literature and in press releases from the research groups involved. Claims about progress in these programs should be checked against primary publications, since I have not verified trial results for any of the three.